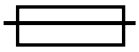
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

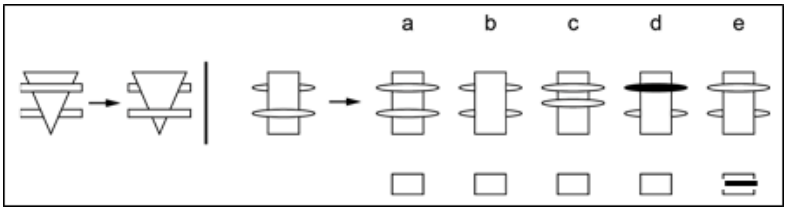
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 25 minutes to complete the paper.
- Mark your answer like this: 
- This paper features a total of 39 questions from 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't answer a question, move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

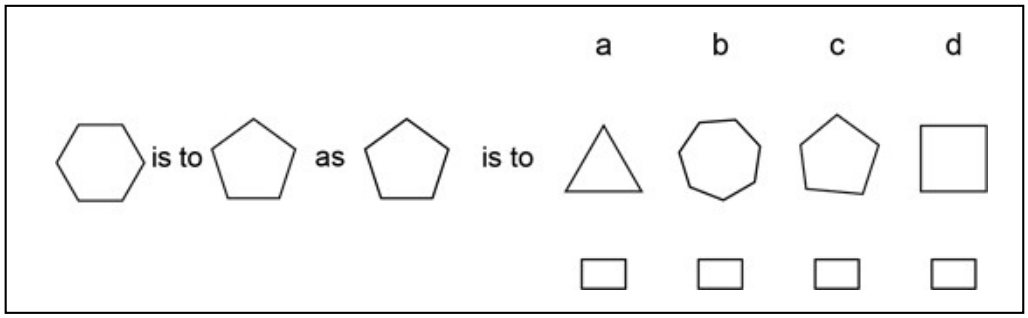
1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

Example



1



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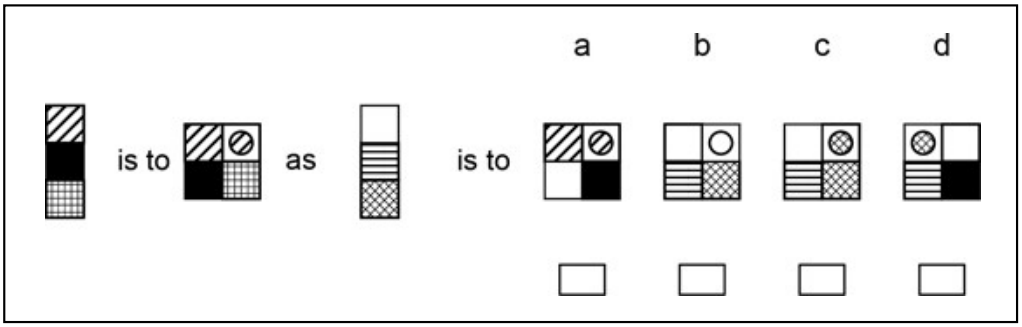

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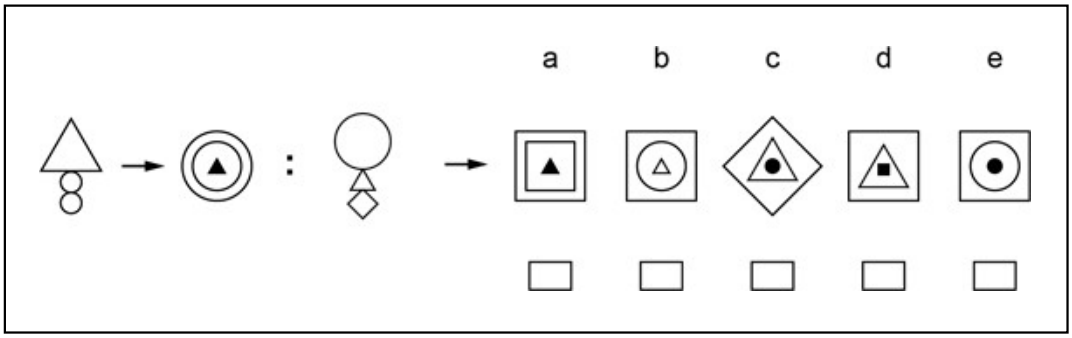
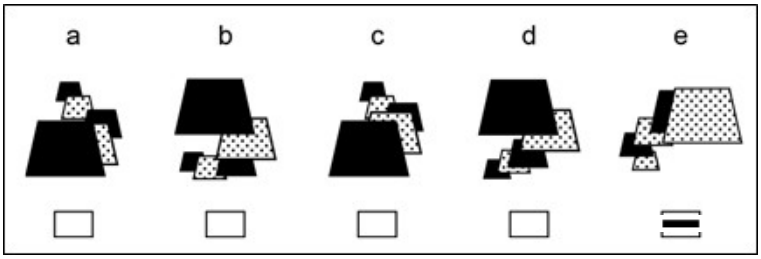


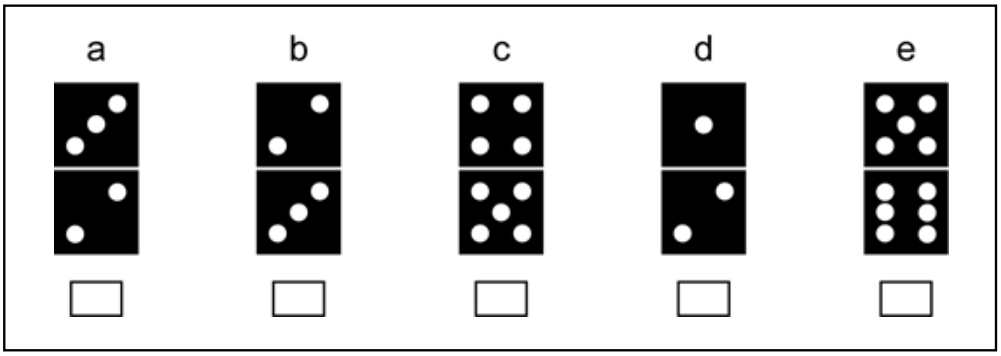
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

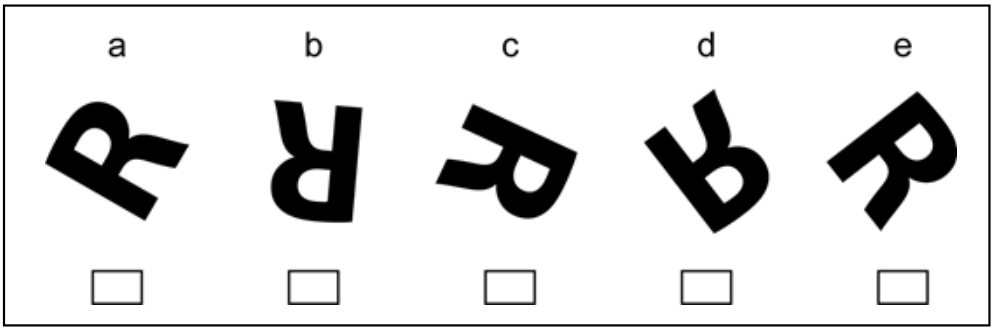
Example



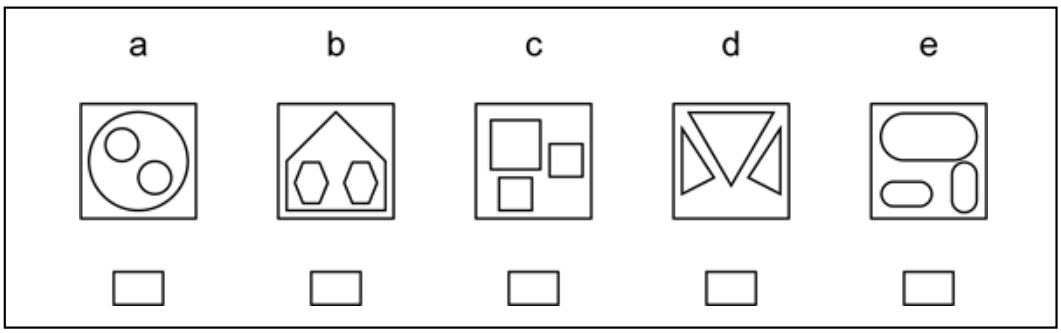
1



2



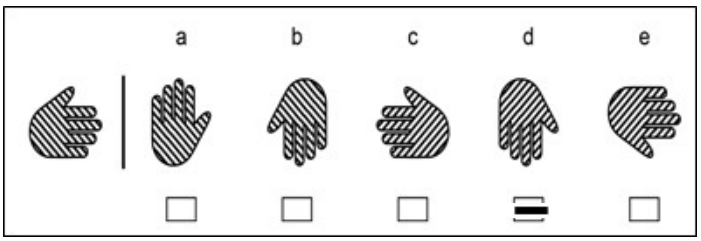
3



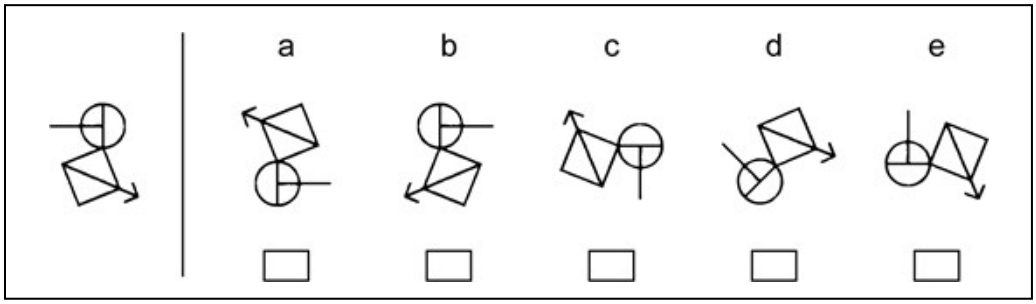
3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

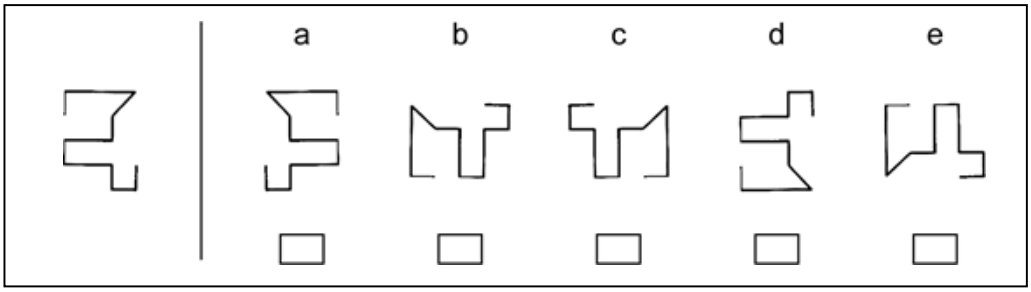
Example



1



2



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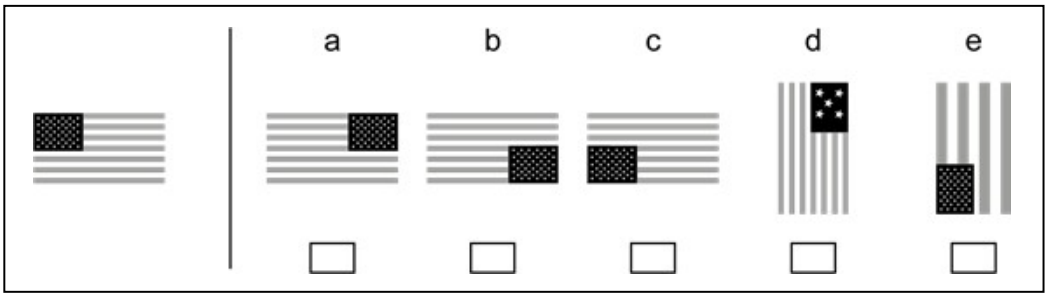
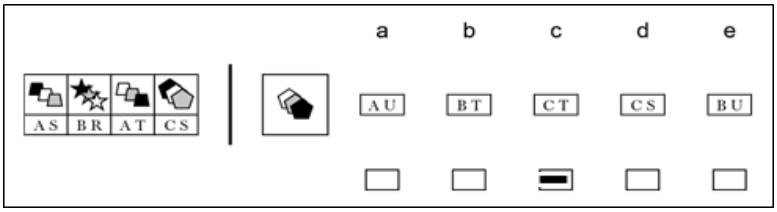


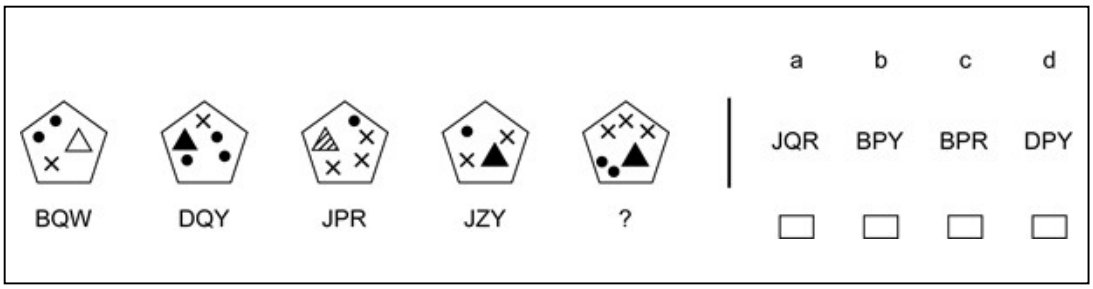
4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

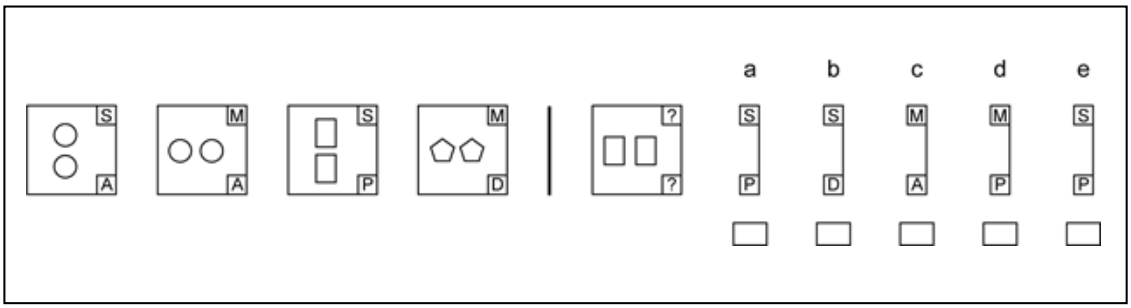
Example



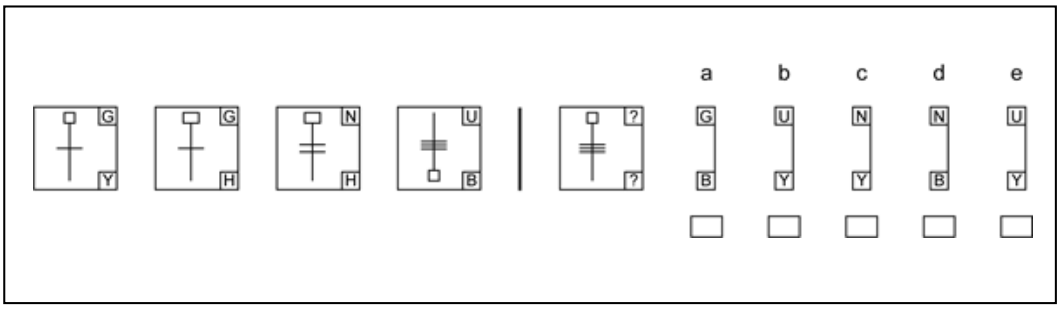
1



2



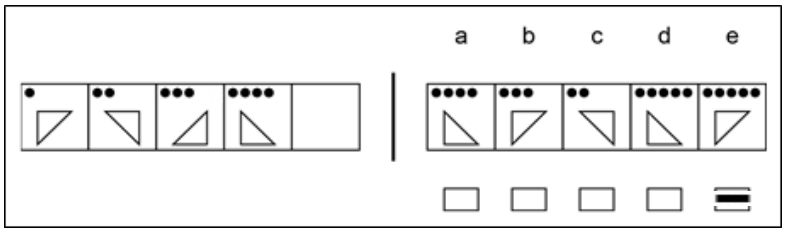
3



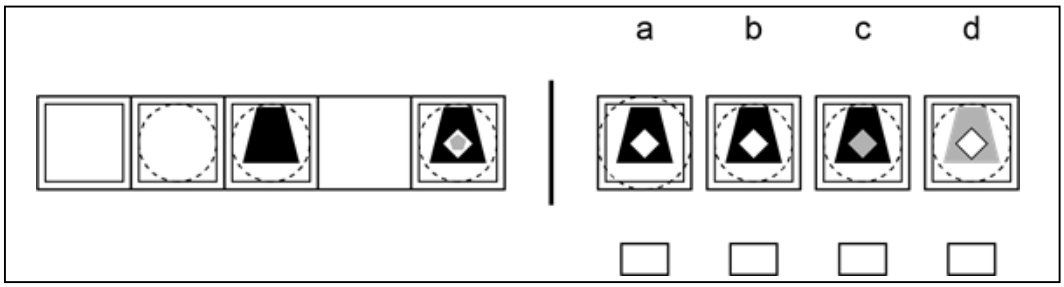
5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

Example



1



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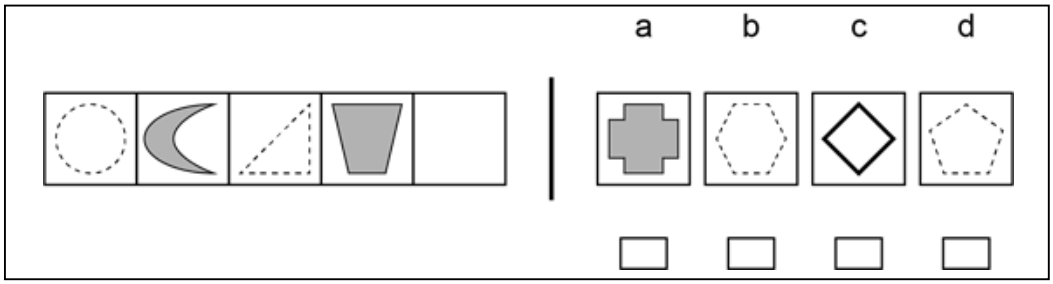
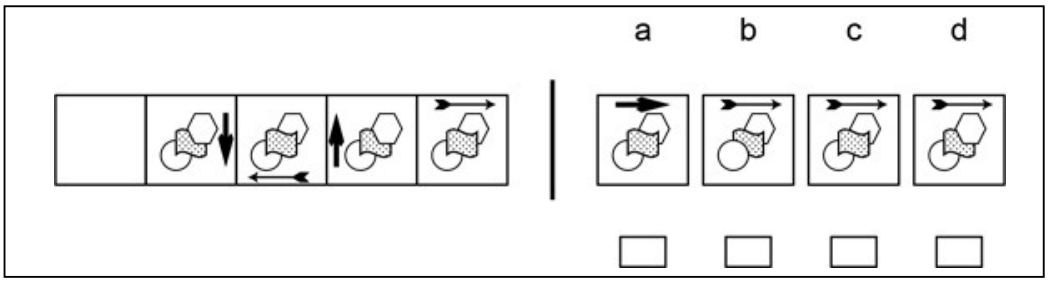


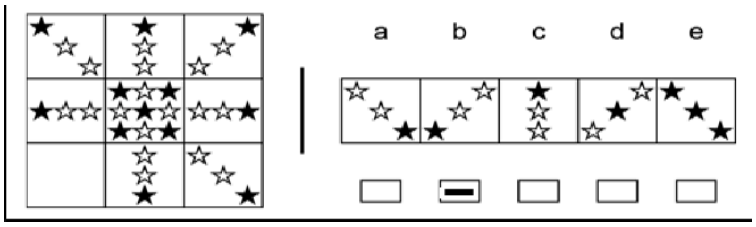
3



6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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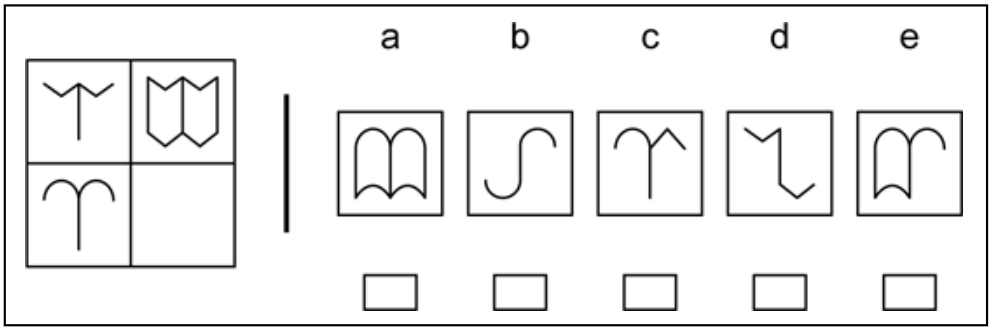

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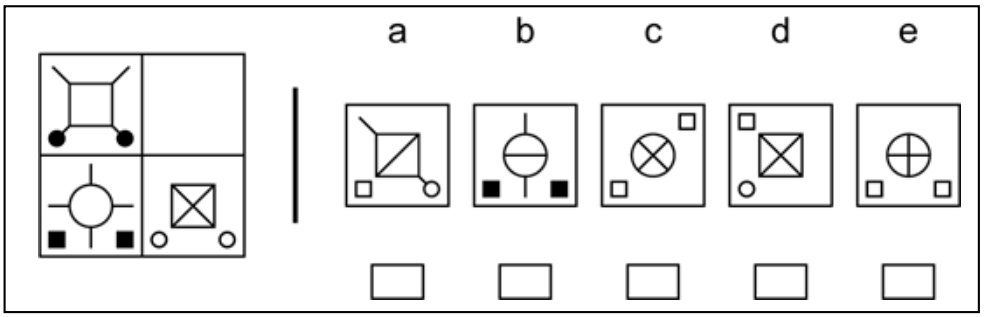

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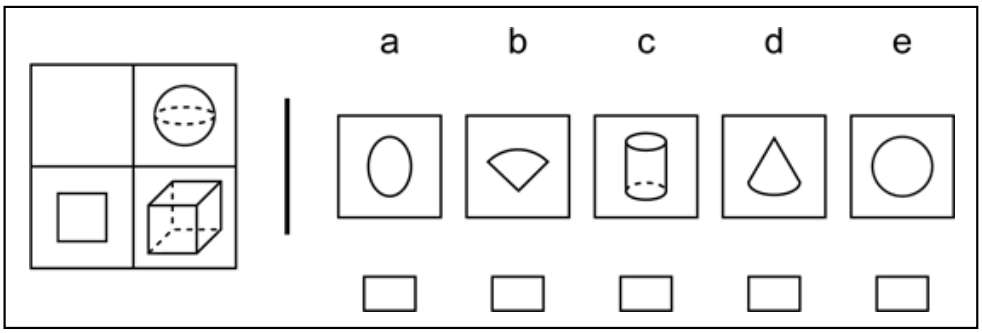
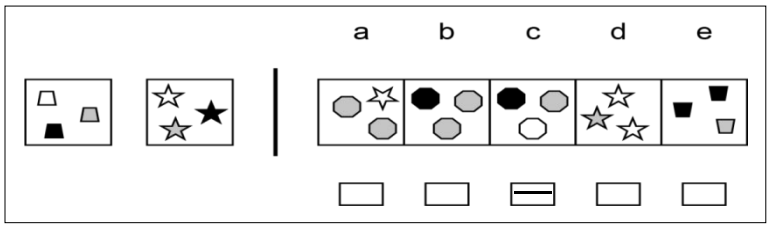


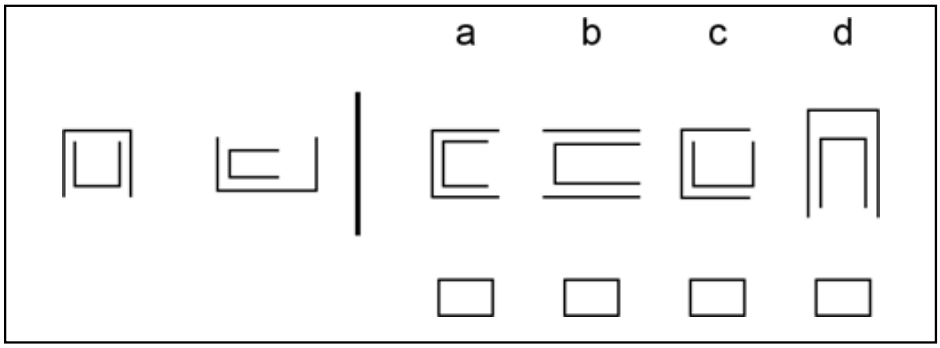
7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

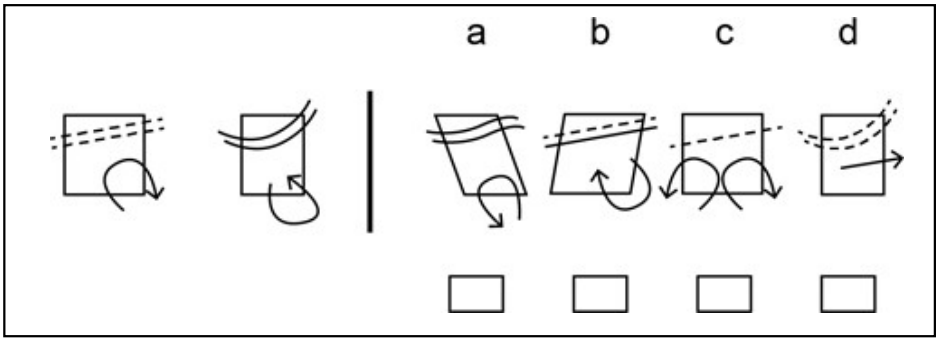
Example



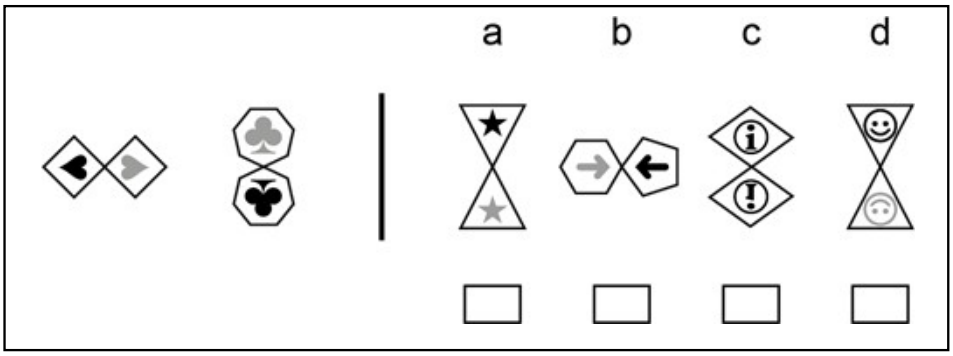
1



2



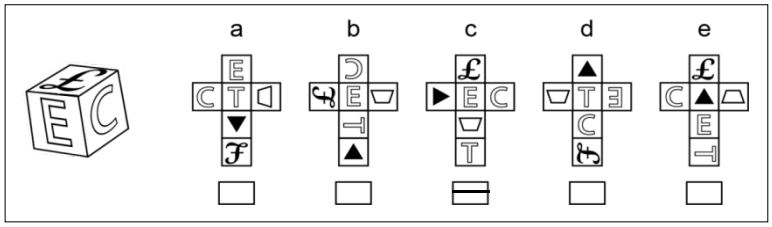
3



8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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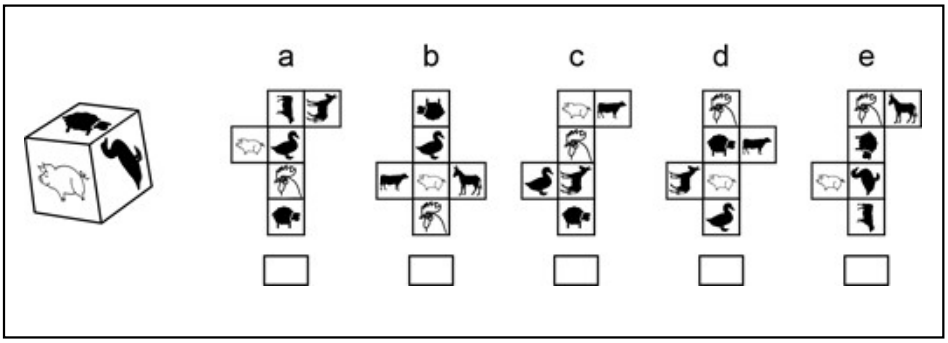
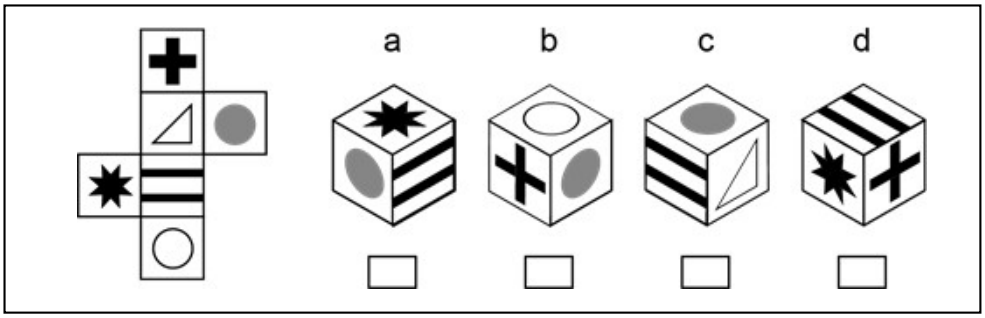
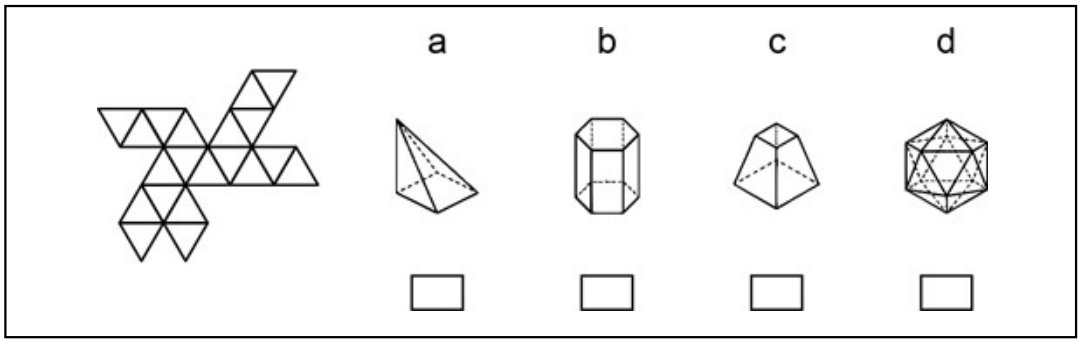


2



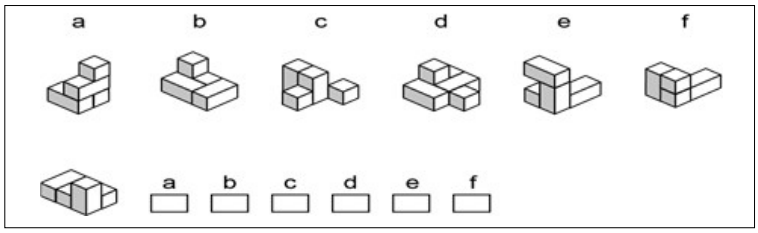
3



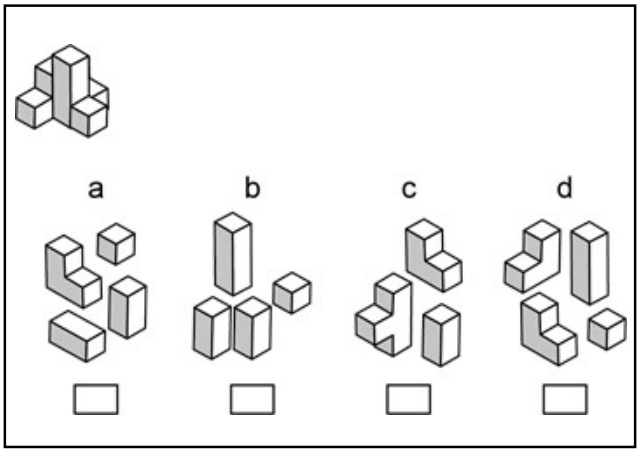
9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

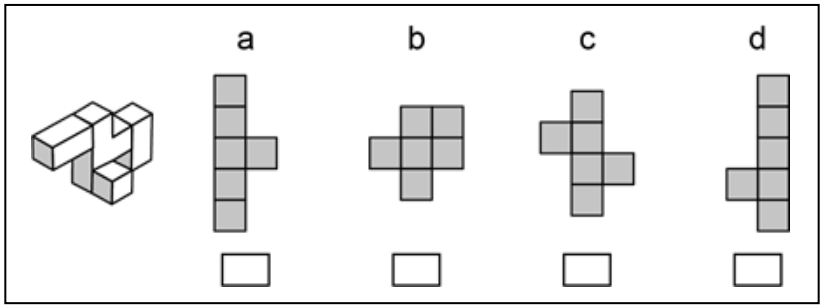
Example



1



2



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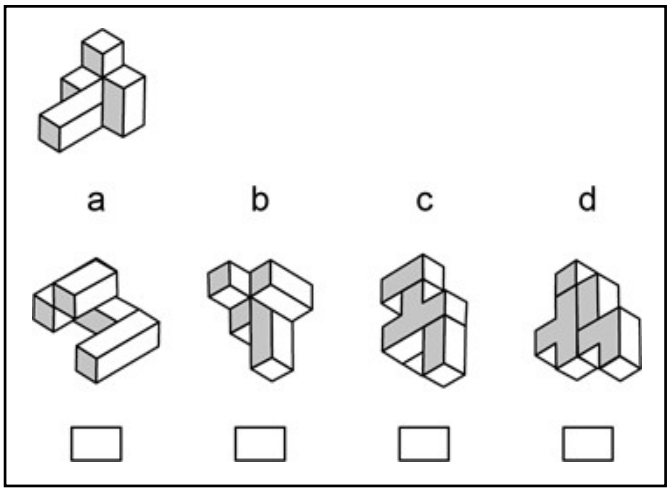
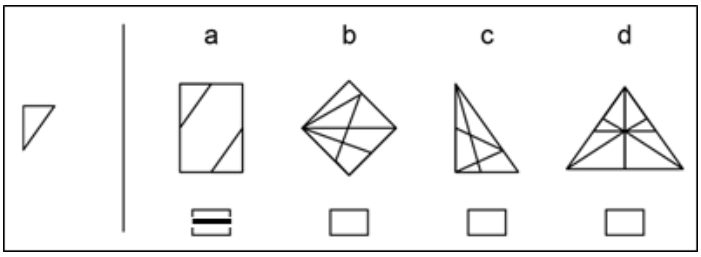


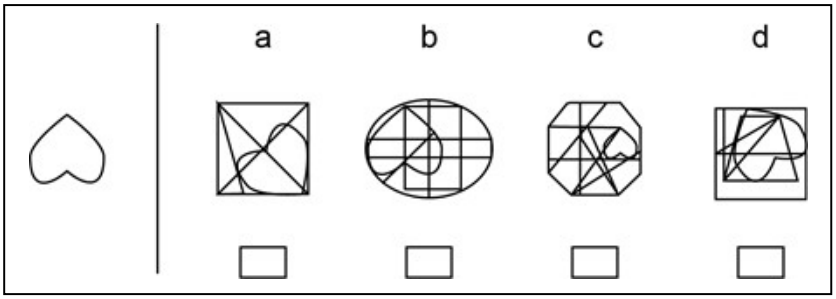
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

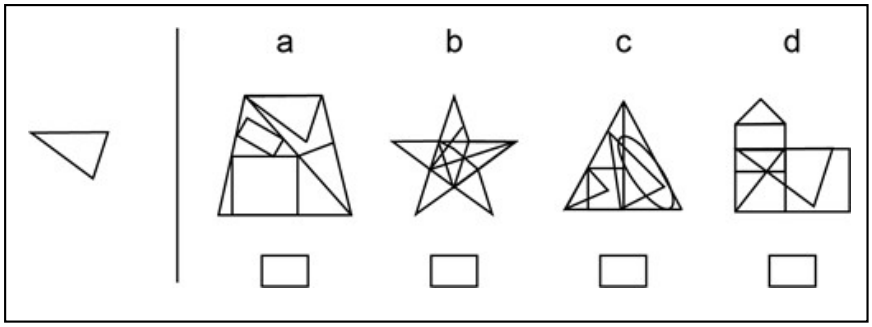
Example



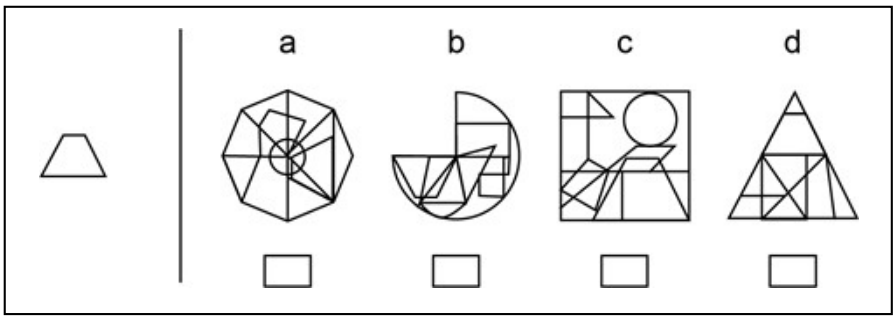
1



2



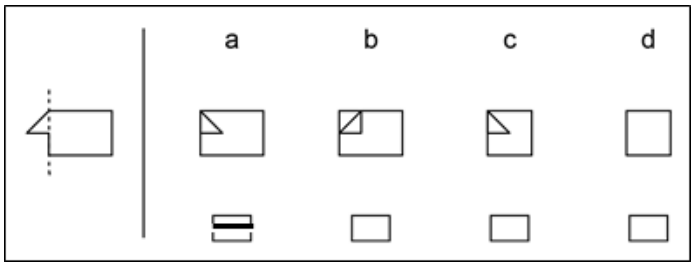
3



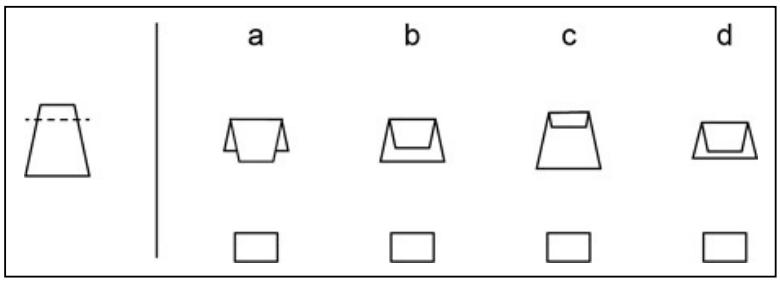
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

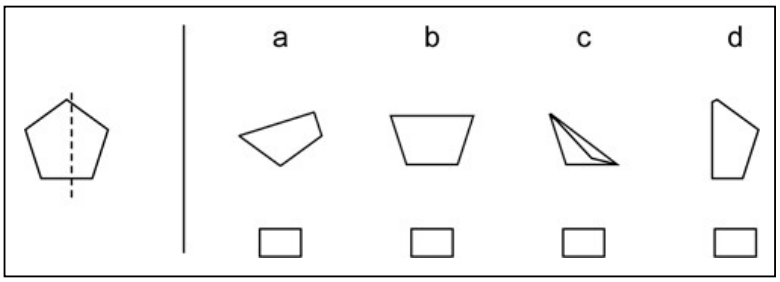
Example



1



2



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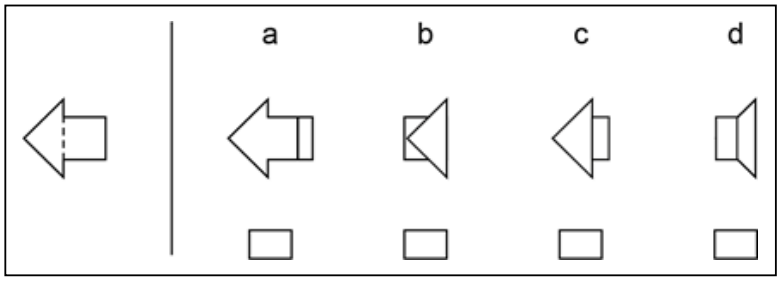
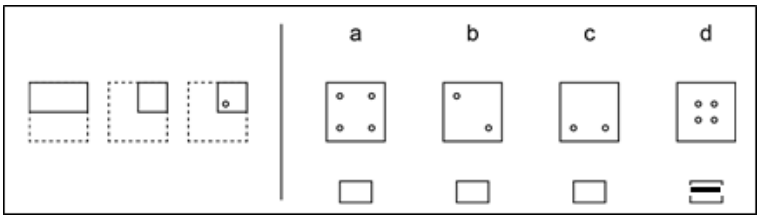


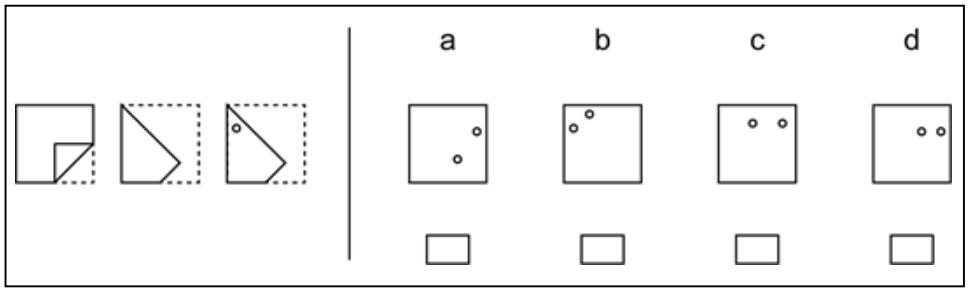
12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

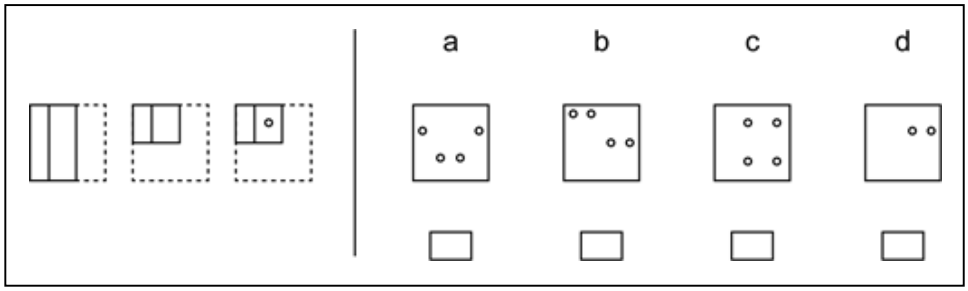
Example



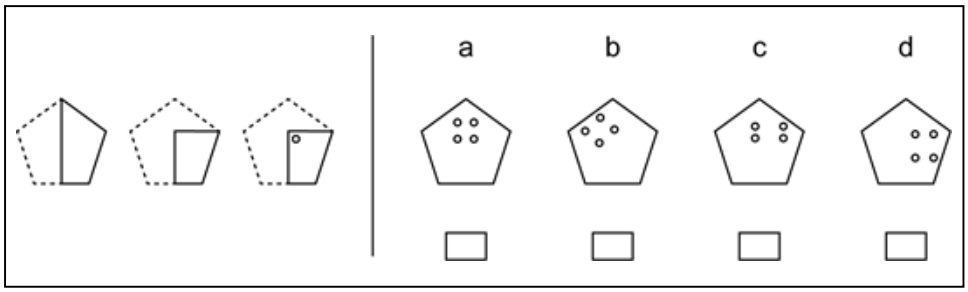
1



2



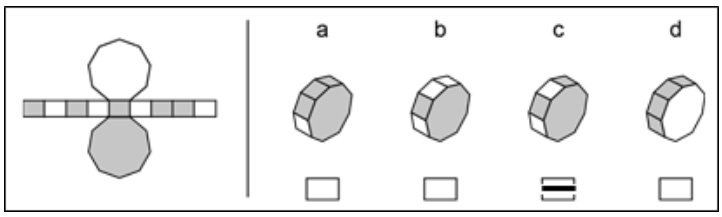
3



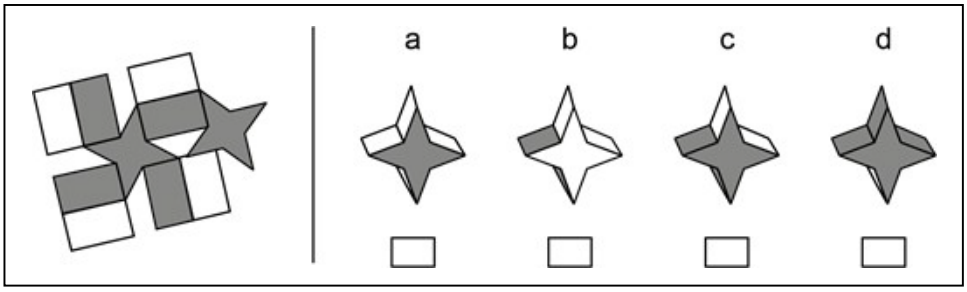
13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

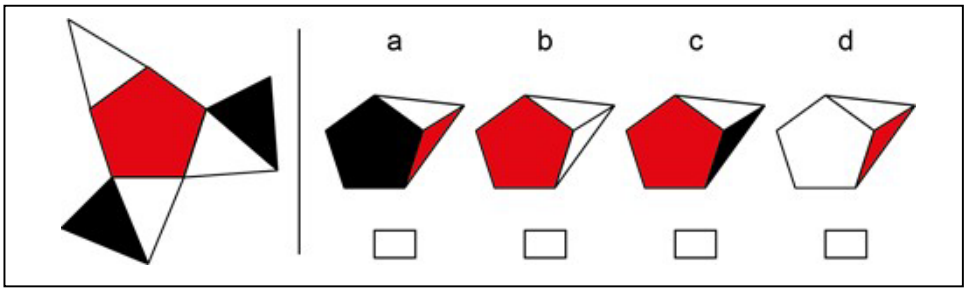
Example



1



2



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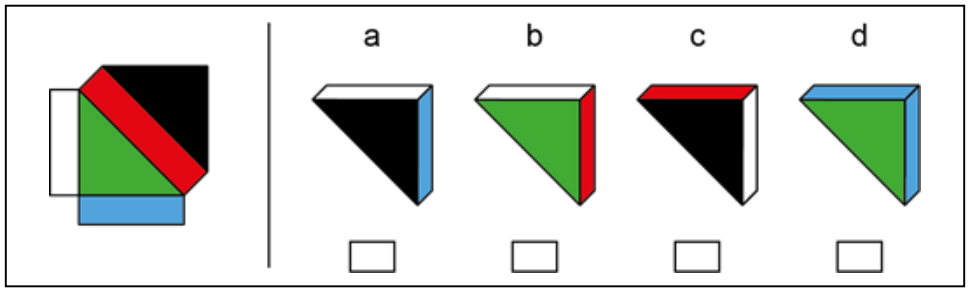


Non-Verbal Reasoning

Like Shapes

1. **D** - (The first shape has one more side than the second shape.)
2. **B** – (The column becomes a square grid with the top square becoming the top left square in the grid, the middle square becoming the bottom left square in the grid, the bottom square becoming the bottom right square in the grid and the top right square of the grid having a circle with the same shading pattern as the top left square.)
3. **C** - (The top large shape becomes the small innermost shape and changes colour from white to black, the second shape down becomes the middle shape and the third shape down becomes the outermost shape.)

Odd One Out

1. **A** – (In all the options the bottom half of the domino has a value one greater than the top half apart from A, which has a greater value in the top half.)
2. **D** – (All answer options are a rotation of the letter R apart from D, which is a reflection.)
3. **B** – (All answer options contain three versions of the same shape in different sizes and orientation apart from B, which contains a pentagon and two hexagons.)

Rotations

1. **A** – (A is a rotation of 180 degrees.)
2. **B** – (B is a rotation of 90 degrees anti-clockwise.)
3. **B** – (B is a rotation of 180 degrees.)

Code Breaker

1. **B** – (You can work out that J stands for there being one black dot as there are two of them. Therefore, B stands for there being two black dots. You can work out that Q stands for there being one X as there are two of them. Therefore, P stands for there being three Xs. Finally, you can work out that Y stands for the triangle being shaded black as there are two of them. Therefore, the final answer is BPY.)

2. **D** - (You can work out that M stands for the shapes being next to each other and S stands for the shapes being stacked vertically as there are two of each. You can then work out that A stands for the shapes being circles as there are two of them. Therefore, P stands for the shapes being rectangles, making the final answer MP.)
3. **B** - (You can work out that G stands for there being one short line as there are two of them. Therefore, U stands for there being four short lines. You can work out that H stands for there being a rectangular shape at the top end as there are two of them. Therefore, Y stands for there being a square shape at the top end, making the final answer UY.)

Missing Sequence

1. **B** - (The sequence builds a picture, with an extra element being added at each stage. Based on the steps you are given, you know that the missing step must enable you to start with a square, dashed circle and black trapezium and finish with a square, dashed circle, black trapezium, white diamond and grey pentagon. Therefore the answer must be B.)
2. **D** - (As you move from left to right the two changes occur. First, the number of sides of the shape increases by one each time. Second, the shape alternates between being white with a dashed border and being grey with a solid border.)
3. **C** - (As you move through the sequence three changes occur at each step. First, the black arrow alternates in style between being thin and thick. Second, the black arrow moves around the square by one side clockwise each step. Third, the shapes alternate their pattern between the middle shape being on top of the other two and the middle shape being between the other two.)

Missing Square

1. **A** - (In each row the beginning of a shape on the left is made in to a complete joined up shape on the right.)
2. **E** - (The squares on the right hand side show the opposite diagonal shape with two changes. First the lines on the outside of the shape move inside the shape. Second, the shapes in the bottom left and right corners change colour from black to white.)

3. **E** - (The squares on the left show the 2D version of the 3D shape on the right.)

Most Like

1. **C** – (Both examples show what would be one four sided shape inside another, each with a different side missing. C is more similar to the examples than A or D because A and D have the same side missing for both shapes.)
2. **A** – (In both examples two lines of the same type go across the main shape. A curved arrow also overlaps the main shape, with both the start and end of the arrow either in or out of the main shape.)
3. **D** – (Both examples show a reflection, with one of the inner shapes black and the other grey. A is incorrect because the inner shape has been reflected incorrectly. B is incorrect because the two outer shapes are different, rather than being a reflection. C is incorrect because both the inner shapes are the same colour.)

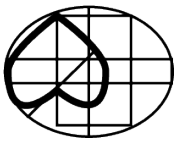
Nets

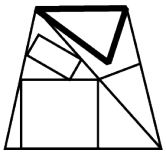
1. **E** - (From the cube you can see that the top of the duck must share an edge with the back of the pig. The only net that satisfies this is E.)
2. **B** – (This is a tricky question. You need to recognise that then the net is folded up completely the white circle will share an edge with the top of the cross and the shaded circle. A cannot be correct as the star and shaded circle are on opposite sides of the net and so cannot share an edge. For the same reason D cannot be correct with respect to the cross and double line. C cannot be correct as the triangle is in the wrong orientation with respect to the shaded circle.)
3. **D** – (Despite initially looking like a very tricky question you will quickly realise that the net cannot possibly make shapes A, B or C because they all have far too few sides.)

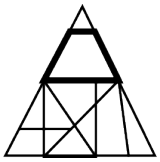
3D Shapes

1. **D** – (You can see from the 3D shape that the correct answer must have a longer block, therefore you can immediately dismiss A and C as incorrect answers. B cannot be correct as there will not be enough blocks to cover all visible parts of the 3D shape.)
2. **A** – (When viewed from above the shape will appear as a column of five squares with one square to the right at the mid-point.)
3. **B** – (B is a rotation of approximately 60 degrees anti-clockwise in the x-axis.)

Hidden Shape

1. **B** – 

2. **A** - 

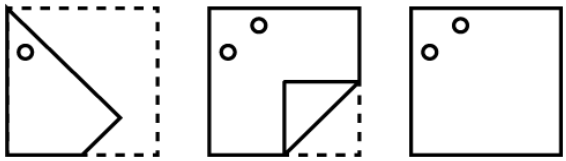
3. **D** – 

Fold Along the Dotted Line

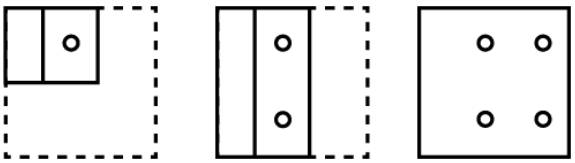
1. **C** – (The top portion of the shape will fold down over the bottom portion.)
2. **D** – (The left portion of the shape will fold over the top of the right portion, covering it completely.)
3. **B** – (The right portion will fold behind the left portion so that part of it remains visible underneath.)

Hole Punch

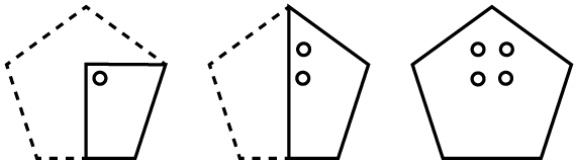
1. **B** -



2. **C** -



3. **A** –



Shaded Nets

1. **C** - (From the net you can see that both star faces must be grey. Therefore B cannot be correct. You can also see that the shading pattern for the rectangular faces must alternate between white and grey. Therefore C must be the correct answer.)
2. **C** - (From the net you can see that the pentagon must be red. Therefore A and D must both be incorrect. B cannot be correct as two white triangular faces do not appear next to one another in the net. Therefore C must be the correct answer.)
3. **A** – (From the net you can see that one of the shorter rectangular faces must be white, while the other shorter rectangular face must be blue. A is the only option that satisfies this.)